Mathematical Foundations of Reinforcement Learning and Online Decision Making Homework 2

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1 Solution to Question 1

Given states $\mathcal{S} = \{1, 2, 3, 4\}$ and actions per state as $\mathcal{A} = \{1, 2\}$. State $S = 4$ is a terminal state with zero cost. Question provides details on transition probabilities $(p_{i,j})$ and cost of a certain action $g(a)$. The problem specified in the question is a case of *deterministic policy*.

1.1 (a)

From the Bellman equation, we know that:

Lemma 1.1.1. *For a shortest path problem with terminal states, the value of starting in state $\mathcal{S}_0 = s$ under a policy $\pi = (\mu_0, \mu_1, \mu_2, \dots)$ is defined as the expected total cost over time:*

$$v^\pi(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_0 = s\right]$$

$$\implies v^\pi(s) = \mathbb{E}\left[g(s, \mathcal{A}_0) + \sum_{k=1}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_0 = s\right]$$

$$\implies v^\pi(s) = \mathbb{E}[g(s, \mathcal{A}_0) \mid \mathcal{S}_0 = s] + \mathbb{E}\left[\sum_{k=1}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_0 = s\right]$$

where $\mathcal{S}_k$ is the state at time step k, $\mathcal{A}_k = \mu_k(\mathcal{S}_k)$ is the action taken at time k, and $g(S_k, A_k)$ is the transition cost.

To evaluate the second term from Lemma 1.1.1, we condition on the state $s' \in \mathcal{S}$ that the system transitions into at step $k = 1$ (using the Law of Total Expectation):

$$\mathbb{E}\left[\sum_{k=1}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_0 = s\right] = \sum_{s' \in \mathcal{S}} \mathbb{P}(\mathcal{S}_1 = s' \mid \mathcal{S}_0 = s, \mathcal{A}_0 = a) \cdot \mathbb{E}\left[\sum_{k=1}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_1 = s'\right]$$

Substituting this back in value equation from Lemma 1.1.1 gives:

$$v^{\pi}(s) = g(s,a) + \sum_{s' \in \mathcal{S}} p_{ss'}(a) \cdot \mathbb{E}\left[\sum_{k=1}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_1 = s' \right]$$

Since given policy is a stationary policy ($\mu_k = \mu$ for all k), the action taken at state s is always $\mu(s)$, and the cost function $g(a)$ in this problem depends only on the action taken ($g(\mu(s))$).

Because the transition probabilities and policy do not change over time due to stationarity property, the expected future cost starting from $\mathcal{S}_1 = s'$ at step 1 is identical to starting from $\mathcal{S}_0 = s'$ at step 0. Therefore:

$$\mathbb{E}\left[\sum_{k=1}^{\infty} g(\mathcal{S}_k, \mathcal{A}_k) \;\middle|\; \mathcal{S}_1 = s' \right] = v^{\mu}(s')$$

Substituting $v^{\mu}(s')$ back into the value equation from Lemma 1.1.1:

Lemma 1.1.2. *We get the stationary Bellman equation as*

$$v^{\mu}(s) = g(\mu(s)) + \sum_{s' \in \mathcal{S}} p_{ss'}(\mu(s)) v^{\mu}(s')$$

Since this equation holds simultaneously for every state s, it converts an infinite-horizon expectation problem into a system of simple linear equations.

As per the question, $v^{\mu}(4) = 0$, $g(1) = -1$ and $g(2) = -2$.

For a stationary policy $\mu_1(s) = 1$, the action taken in all states is $a = 1$ and the cost is $g(1) = -1$. The transition probabilities for $a = 1$ are:

$$p_{ii}(1) = 0.9 \quad \text{for} \quad i \in \{1,2,3\}$$

$$p_{i,i+1}(1) = 0.1 \quad \text{for} \quad i \in \{1,2,3\}$$

And for any other p, it is 0.

Now computing the value v^{μ_1} through above Bellman equation is:

$$\implies v^{\mu_1} = \begin{bmatrix} v(1) = -1 + 0.9v(1) + 0.1v(2) \\ v(2) = -1 + 0.9v(2) + 0.1v(3) \\ v(3) = -1 + 0.9v(3) + 0.1v(4) \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_1} = \begin{bmatrix} v(1) = -1 + 0.9v(1) + 0.1v(2) \\ v(2) = -1 + 0.9v(2) + 0.1v(3) \\ v(3) = -10 \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_1} = \begin{bmatrix} v(1) = -1 + 0.9v(1) + 0.1v(2) \\ v(2) = -20 \\ v(3) = -10 \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_1} = \begin{bmatrix} -30 \\ -20 \\ -10 \\ 0 \end{bmatrix}$$

For a stationary policy $\mu_2(s) = 2$, the action taken in all states is $a = 2$ and the cost is $g(1) = -2$. The transition probabilities for $a = 2$ are:

$$p_{11}(2) = p_{12}(2) = 0.5$$

$$p_{i,i+1}(2) = p_{i,i-1}(2) = 0.5 \quad \text{for} \quad i \in \{2, 3\}$$

And for any other p, it is 0.

Now computing the value v^{μ_2} through above Bellman equation is:

$$\implies v^{\mu_2} = \begin{bmatrix} v(1) = -2 + 0.5v(1) + 0.5v(2) \\ v(2) = -2 + 0.5v(1) + 0.5v(3) \\ v(3) = -2 + 0.5v(2) + 0.5v(4) \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_2} = \begin{bmatrix} v(1) = -4 + v(2) \\ v(2) = -2 + 0.5v(1) + 0.5v(3) \\ v(3) = -2 + 0.5v(2) \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_2} = \begin{bmatrix} v(1) = -4 + v(2) \\ v(2) = -2 + 0.5\left(v(2) - 4\right) + 0.5\left(0.5v(2) - 2\right) \\ v(3) = -2 + 0.5v(2) \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_2} = \begin{bmatrix} v(1) = -4 + v(2) \\ v(2) = -20 \\ v(3) = -2 + 0.5v(2) \\ v(4) = 0 \end{bmatrix}$$

$$\implies v^{\mu_2} = \begin{bmatrix} v(1) = -24 \\ v(2) = -20 \\ v(3) = -12 \\ v(4) = 0 \end{bmatrix}$$

1.2 (b)

Lemma 1.2.1. *Considering the Bellman Optimality Equation for this scenario, we get:*

$$v^*(s) = \min/\max_{a \in \mathcal{A}} \left\{ g(a) + \sum_{s' \in \mathcal{S}} p_{ss'}(a) v^*(s') \right\}$$

For our problem of shortest path, we get the Bellman optimality equation as:

$$v^*(s) = \min_{a\in\{1,2\}} \left\{ g(a) + \sum_{s'\in\{1,2,3\}} p_{ss'}(a) v^*(s') \right\}$$

with $v*(4) = 0$, state 4 being cost-free and $p_{4,s \text{ for} s\in\mathcal{S}} = 1$.

Now, for each optimal value for $s \in \{1, 2, 3\}$, we solve it one by one.

For $s = 1$:

$$v^*(1) = \min\{-1 + 0.9v^*(1) + 0.1v^*(2), -2 + 0.5v^*(1) + 0.5v^*(2)\}$$
$$= \min\{-10 + v^*(2), -4 + v^*(2)\}$$

Here, for any given value of $v^*(2)$, $-10 + v^*(2) < -4 + v^*(2)$. So the $v^*(1) = \min\{-10 + v^*(2), -4 + v^*(2)\} = -10 + v^*(2)$ for $\mu^*(1) = 1$.

For $s = 2$:

$$v^*(2) = \min\{-1 + 0.9v^*(2) + 0.1v^*(3), -2 + 0.5v^*(1) + 0.5v^*(3)\}$$
$$= \min\{-10 + v^*(3), -2 + 0.5v^*(3) + 0.5v^*(1)\}$$

Substituting $v^*(1)$ above, we get:

$$v^*(2) = \min\{-10 + v^*(3), -14 + v^*(3)\}$$

Here, for any given value of $v^*(2)$, $-14 + v^*(3) < -10 + v^*(3)$. So the $v^*(2) = \min\{-10 + v^*(3), -14 + v^*(3)\} = -14 + v^*(3)$ for $\mu^*(2) = 2$.

For $s = 3$:

$$v^*(3) = \min\{-1 + 0.9v^*(3) + 0.1v^*(4), -2 + 0.5v^*(2) + 0.5v^*(4)\}$$
$$= \min\{-10, -2 + 0.5v^*(3) + 0.5v^*(1)\}$$

Substituting $v^*(2)$ above, we get:

$$v^*(3) = \min\{-10, -18\}$$

So the $v^*(3) = \min\{-10, -18\} = -18$ for $\mu^*(3) = 2$.

Substituting the $v^*(3)$ for $v^*(2)$, we get $v^*(2) = -32$. Substituting the $v^*(2)$ for $v^*(1)$, we get $v^*(1) = -42$.

So, we have the final optimal policy:

$$\mu^*(s) = \begin{cases} 1 & \text{if } s = 1 \\ 2 & \text{if } s = 2 \\ 2 & \text{if } s = 3 \\ \text{not defined} & \text{if } s = 4 \end{cases}$$

For optimal value:

$$v^*(s) = \begin{cases} -42 & \text{if } s = 1 \\ -32 & \text{if } s = 2 \\ -18 & \text{if } s = 3 \\ 0 & \text{if } s = 4 \end{cases}$$

1.3 (c)

We already have the cost, actions and transition probabilities from above.

Since state $s = 4$ is terminal with $v(4) = 0$, the non-terminal value vector can be represented in $\mathbb{R}^3$ as $\hat{v} = \begin{bmatrix} v(1) & v(2) & v(3) \end{bmatrix}^T$.

The optimal Bellman operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ for shortest path problem is defined component-wise for $s \in \{1, 2, 3\}$ as:

$$[T\hat{v}](s) = \min_{a \in \{1,2\}} \left\{ g(a) + \sum_{s' \in \{1,2,3\}} P(s' \mid s, a)\hat{v}(s') \right\}$$

We assume there exists some norm $\|\cdot\|$ and a parameter $\beta \in [0, 1)$ such that for all $\hat{u}, \hat{v} \in \mathbb{R}^3$:

$$\|T\hat{u} - T\hat{v}\| \leq \beta \|\hat{u} - \hat{v}\|$$

For any two value vectors $\hat{u}, \hat{v} \in \mathbb{R}^3$ and any state $s \in \{1, 2, 3\}$, the non-expansiveness of the min operator implies:

$$|(T\hat{u})(s) - (T\hat{v})(s)| \leq \max_{a \in \{1,2\}} \sum_{s' \in \{1,2,3\}} P(s' \mid s, a)|\hat{u}(s') - \hat{v}(s')|$$

Considering a generic weighted norm (which is also a $||\cdot||$) on $\mathbb{R}^3$ parameterized by a strictly positive vector $w = (w_1, w_2, w_3)^T > 0$:

$$\|\hat{x}\|_w = \max_{s \in \{1,2,3\}} \frac{|\hat{x}(s)|}{w_s}$$

If $w = (1, 1, 1)^T$, this reduces to the standard unweighted norm.

From the definition of $\|\cdot\|_w$, we have $|\hat{u}(s') - \hat{v}(s')| \leq w_{s'}\|\hat{u} - \hat{v}\|_w$ for all s'. Substituting this generic $w_{s'}$ into the inequality gives:

$$|(T\hat{u})(s) - (T\hat{v})(s)| \leq \left(\max_{a \in \{1,2\}} \sum_{s' \in \{1,2,3\}} P(s' \mid s, a) w_{s'} \right) \|\hat{u} - \hat{v}\|_w$$

Dividing both sides by w_s:

$$\frac{|(T\hat{u})(s) - (T\hat{v})(s)|}{w_s} \leq \left(\frac{1}{w_s} \max_{a \in \{1,2\}} \sum_{s' \in \{1,2,3\}} P(s' \mid s, a) w_{s'} \right) \|\hat{u} - \hat{v}\|_w$$

Taking the maximum over $a \in \{1, 2\}$, T is a β-contraction in $||\cdot||$ ($\|T\hat{u} - T\hat{v}\|_w \leq \beta\|\hat{u} - \hat{v}\|_w$) if and only if there exist positive scalars $w_1, w_2, w_3 > 0$ (vector norm requirements) and a parameter $\beta \in [0, 1)$ satisfying the system of inequalities:

$$\frac{1}{w_s} \sum_{s' \in \{1,2,3\}} P(s' \mid s, a) w_{s'} \leq \beta \quad \forall s \in \{1, 2, 3\},\ \forall a \in \{1, 2\}$$

Now, if we set the standard unweighted norm ($w_1 = w_2 = w_3 = 1$). Setting $w = (1, 1, 1)^T$ into the system of inequalities for $a = 1$:

State 1 ($a = 1$): $P_{11}(1) \cdot 1 + P_{12}(1) \cdot 1 = 0.9 + 0.1 = 1.0 \leq \beta$

State 2 ($a = 1$): $P_{22}(1) \cdot 1 + P_{23}(1) \cdot 1 = 0.9 + 0.1 = 1.0 \leq \beta$

This requires $\beta \geq 1.0$, which violates the strict contraction requirement $\beta \in [0, 1)$ given in the question. Thus, a standard unweighted norm cannot establish contraction.

To guarantee $\beta \in [0, 1)$, we must solve the system of 6 linear inequalities explicitly for (w_1, w_2, w_3) and β:

$$s = 1, a = 1 : 0.9w_1 + 0.1w_2 \leq \beta w_1 \implies 0.1w_2 \leq (\beta - 0.9)w_1 \tag{1.3.1}$$

$$s = 2, a = 1 : 0.9w_2 + 0.1w_3 \leq \beta w_2 \implies 0.1w_3 \leq (\beta - 0.9)w_2 \tag{1.3.2}$$

$$s = 3, a = 1 : 0.9w_3 \leq \beta w_3 \implies \beta \geq 0.9 \tag{1.3.3}$$

$$s = 1, a = 2 : 0.5w_1 + 0.5w_2 \leq \beta w_1 \implies 0.5w_2 \leq (\beta - 0.5)w_1 \tag{1.3.4}$$

$$s = 2, a = 2 : 0.5w_1 + 0.5w_3 \leq \beta w_2 \tag{1.3.5}$$

$$s = 3, a = 2 : 0.5w_2 \leq \beta w_3 \tag{1.3.6}$$

Let's choose $0.9 \leq \beta = 0.95 < 1$, satisfying Equation 1.3.3.

Let's choose $w_1 = 0.5 > 0$, being strictly positive.

So, from Equation 1.3.1, we get $w_2 \leq 0.25$.

Let's take $w_2 = 0.2 \leq 0.25$.

Using $w_2 = 0.2$ in Equation 1.3.6, we get $w_3 \geq \frac{1}{95} \approx 0.01$.

So, we have $\beta = 0.95$, and $w_s = (0.5, 0.2, 0.01)^T$ which should satisfy the T's $\beta-$contractivity in $||\cdot||$.

2 Solution to Question 2

We are given a Markov Reward Process with states $\mathcal{S} = \{0, 1, \ldots, S\}$, transition matrix P and reward function $R : \mathcal{S} \to \mathbb{R}$.

State 0 is absorbing, with $R(0) = 0$ and $P(0, 0) = 1$ and $P^k(i, 0) > 0 \; \forall i \in \mathcal{S}$ and some $k \in \mathbb{N}$.

Given the value function for the shortest path problem:

$$v(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} R_k | S_0 = s\right]$$

v and r can be represented as column vectors $v, r \in \mathbb{R}^{S+1}$ where

$$v = (v(0), v(1), \ldots, v(S)) \quad \text{and} \quad r = (r(0), r(1), \ldots, r(S))$$

If absorbing states are excluded from v and r, then we write:

$$\hat{v} : S_+ \to \mathbb{R}^S \quad \text{and} \quad \hat{r} : S_+ \to \mathbb{R}^S, \quad \text{where} \quad \mathcal{S}_+ = S \setminus \{0\}$$

The steps of computing $v(s)$ is done by solving system of linear equations.

2.1 (a)

We need to show that the transition matrix P has the below structure

$$P = \begin{bmatrix} 1 & \mathbf{0} \\ P_1 & P_2 \end{bmatrix}$$

where $P_1 \in \mathbb{R}^{S\times 1}$, $P_2 \in \mathbb{R}^{S\times S}$, and $\mathbf{0} = [0, \ldots, 0] \in \mathbb{R}^{1\times S}$.
As per the question, $\mathcal{S} = \{0, 1, \ldots, S\}$.
If we partition $\mathcal{S}$ into two disjoint sets:
The absorbing state set: $\{0\}$
The non-absorbing states: $\mathcal{S}_+ = \{1, \ldots, S\} = \mathcal{S} \setminus \{0\}$.
Since the states in $\mathcal{S}$ are ordered sequentially as $(0, 1, 2, \ldots, S)$, the transition matrix $P \in \mathbb{R}^{(S+1)\times(S+1)}$ can be partitioned into four block sub-matrices corresponding to transitions between $\{0\}$ and $\mathcal{S}_+$:

$$P = \begin{bmatrix} P(0,0) & P(0,\mathcal{S}_+) \\ P(\mathcal{S}_+,0) & P(\mathcal{S}_+,\mathcal{S}_+) \end{bmatrix}$$

where,

- $P(0,0) \in \mathbb{R}^{1\times 1}$ represents the transition probability from state 0 to state 0
- $P(0,\mathcal{S}_+) \in \mathbb{R}^{1\times S}$ represents the row vector of transition probabilities from state 0 to any state in $\mathcal{S}_+$
- $P(\mathcal{S}_+,0) \in \mathbb{R}^{S\times 1}$ represents the column vector of transition probabilities from any state in $\mathcal{S}_+$ to state 0
- $P(\mathcal{S}_+,\mathcal{S}_+) \in \mathbb{R}^{S\times S}$ represents the sub-matrix of transition probabilities between states in $\mathcal{S}_+$

As per the question, $P(0,0) = 1$.
From the property of P in markov chains, for any row s in a transition probability matrix P, the sum of probabilities of transitioning from state s to all possible next states $s' \in \mathcal{S}$ must equal 1:

$$\sum_{s'\in\mathcal{S}} P(s,s') = 1$$

Also, every transition probability $P(s, s')$ must be non-negative:

$$P(s, s') \geq 0 \quad \forall s, s' \in \mathcal{S}$$

Given, $P(0, 0) = 1$, applying the total probability axiom to row 0 (where $\mathcal{S} = \{0\} \cup \mathcal{S}_+$):

$$\sum_{s' \in \mathcal{S}} P(0, s') = P(0, 0) + \sum_{s' \in \mathcal{S}_+} P(0, s') = 1$$

Substituting $P(0, 0) = 1$:

$$1 + \sum_{s' \in \mathcal{S}_+} P(0, s') = 1 \implies \sum_{s' \in \mathcal{S}_+} P(0, s') = 0$$

Since $P(0, s') \geq 0$ for all $s' \in \mathcal{S}_+$, a sum of non-negative real numbers can only equal zero if every individual term is zero:

$$P(0, s') = 0 \quad \forall s' \in \mathcal{S}_+$$

Thus, the row vector $P(0, \mathcal{S}_+)$ consists entirely of zeros: $\mathbf{0} = [0, \ldots, 0] \in \mathbb{R}^{1 \times S}$.

By placing the derived row $[1 \quad \mathbf{0}]$ in the first row and matching the lower sub-matrices to their given definitions in the problem, $P_1 \in \mathbb{R}^{S \times 1}$ and $P_2 \in \mathbb{R}^{S \times S}$ (where $P_1(s) = P(s, 0)$ and $P_2(s, s') = P(s, s')$ for $s, s' \in \mathcal{S}_+$), we directly obtain:

$$P = \begin{bmatrix} 1 & \mathbf{0} \\ P_1 & P_2 \end{bmatrix}$$

2.2 (b)

We need to prove that $I - P_2$ is nonsingular, where $I \in \mathbb{R}^{S \times S}$ is the identity matrix and $P_2 \in \mathbb{R}^{S \times S}$ is the sub-matrix of the transition matrix P corresponding to the non-absorbing states $\mathcal{S}_+ = \{1, \ldots, S\}$.

We already have the transition matrix $P \in \mathbb{R}^{(S+1) \times (S+1)}$ over the state space $\mathcal{S} = \{0, 1, \ldots, S\}$ is partitioned as:

$$P = \begin{bmatrix} 1 & \mathbf{0}^T \\ P_1 & P_2 \end{bmatrix}$$

Because P is a valid row-stochastic transition matrix, the sum of each row is 1:

$$\sum_{j \in \mathcal{S}} P(i, j) = 1 \quad \implies \quad P_1(i) + \sum_{j \in \mathcal{S}_+} P_2(i, j) = 1, \quad \forall i \in \mathcal{S}_+$$

Since $P_1(i) \geq 0$ as per the given question (if $k = 1 \in \mathbb{N}$), it follows that for all $i \in \mathcal{S}_+$:

$$\sum_{j=1}^{S} P_2(i, j) \leq 1$$

Computing the powers of matrix P via block multiplication yields:

$$P^k = \begin{bmatrix} 1 & \mathbf{0}^T \\ P_1^k & P_2^k \end{bmatrix} \quad \text{for any } k \geq 1$$

We know that

$$0 < \sum_{j=1}^{S} P_2(i,j) \leq 1 \quad \text{for all } i \in \mathcal{S}_+$$

Lemma 2.2.1. *We say the square matrix A is singular if $|A| = 0$, and non-singular or invertible if $|A| \neq 0$. And from linear system of equations, we know that a homogeneous systems: $Ax = 0$ has non-trivial solutions if and only if $|A| = 0$.*

So, by the above lemma, we can show that $I - P_2$ is non-singular by showing that its kernel (null space) contains only the zero vector $\mathbf{0}$ [i.e., if $(I - P_2)x = \mathbf{0}$, then $x = \mathbf{0}$].

Suppose there exists a vector $x = [x_1, x_2, \ldots, x_S]^T \in \mathbb{R}^S$ such that:

$$(I - P_2)x = \mathbf{0} \quad \implies \quad x = P_2 x$$

Written entry-wise for each state $i \in \mathcal{S}_+$:

$$x_i = \sum_{j=1}^{S} P_2(i,j) x_j$$

Let i^* be a state in $\mathcal{S}_+$ where $|x_i|$ achieves its maximum value, and let $M = |x_{i^*}| = \max_{i \in \mathcal{S}_+} |x_i|$.

By applying the triangle inequality to the entry-wise equation for state i^*:

$$M = |x_{i^*}| = \left| \sum_{j=1}^{S} P_2(i^*, j) x_j \right| \leq \sum_{j=1}^{S} P_2(i^*, j) |x_j| \leq \left(\sum_{j=1}^{S} P_2(i^*, j) \right) M$$

From the problem, we know that state 0 is accessible from every state $i \in \mathcal{S}_+$ through $P^k(i, 0) > 0$.

Therefore, there must exist at least one state $c \in \mathcal{S}_+$ that has a direct non-zero probability of transitioning to state 0, meaning $P(c, 0) > 0$.

For this state c, the row sum of P_2 is strictly less than 1:

$$\sum_{j=1}^{S} P_2(c,j) = 1 - P(c,0) < 1$$

If we evaluate the inequality at this specific state c:

$$|x_c| \leq \sum_{j=1}^{S} P_2(c,j) |x_j| \leq \left(\sum_{j=1}^{S} P_2(c,j) \right) M < M \quad (\text{unless } M = 0)$$

Since $M = \max_{i \in \mathcal{S}_+} |x_i|$, every state j that transitions into state 0 (or transitions into states that eventually reach state 0) forces $|x_j| < M$ unless $M = 0$. Because every state in $\mathcal{S}_+$ can reach state 0 in finite steps ($k \in \mathbb{N}$), through inequality, we get:

$$M = 0$$

Since $M = \max_{i \in \mathcal{S}_+} |x_i| = 0$, it must be that $x = \mathbf{0}$.

Because the only solution to $(I - P_2)x = \mathbf{0}$ is $x = \mathbf{0}$, the matrix $I - P_2$ is non-singular.

2.3 (c)

Given the value function definition in the problem:

$$v(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} R_k \;\middle|\; S_0 = s\right]$$

For the absorbing state $s = 0$, we are given two conditions:

$R(0) = 0$, meaning $R_k = 0$ whenever $S_k = 0$.

$P(0, 0) = 1$, meaning if $S_0 = 0$, then $S_k = 0$ almost surely for all $k \geq 0$.

Therefore, given $S_0 = 0$, the reward at every time step $k \geq 0$ is deterministically zero ($R_k = 0$). Evaluating the expectation directly:

$$v(0) = \mathbb{E}\left[\sum_{k=0}^{\infty} R_k \;\middle|\; S_0 = 0\right] = \sum_{k=0}^{\infty} \mathbb{E}[R_k \mid S_0 = 0] = \sum_{k=0}^{\infty} 0 = 0$$

For any state $s \in \mathcal{S}_+ = \{1, 2, \ldots, S\}$, expanding the series by splitting the initial reward $R_0 = R(s)$ from the remaining summation:

$$v(s) = \mathbb{E}\left[R_0 + \sum_{k=1}^{\infty} R_k \;\middle|\; S_0 = s\right] = R(s) + \mathbb{E}\left[\sum_{k=1}^{\infty} R_k \;\middle|\; S_0 = s\right]$$

Using the Law of Total Expectation by using the first step state $S_1 = s'$:

$$\mathbb{E}\left[\sum_{k=1}^{\infty} R_k \;\middle|\; S_0 = s\right] = \sum_{s' \in \mathcal{S}} \mathbb{P}(S_1 = s' \mid S_0 = s)\, \mathbb{E}\left[\sum_{k=1}^{\infty} R_k \;\middle|\; S_1 = s'\right]$$

Using the time-homogeneity and Markov property of the process:

$$\mathbb{E}\left[\sum_{k=1}^{\infty} R_k \;\middle|\; S_1 = s'\right] = \mathbb{E}\left[\sum_{k=0}^{\infty} R_k \;\middle|\; S_0 = s'\right] = v(s')$$

Substituting this back into the previous $v(s)$ gives:

$$v(s) = R(s) + \sum_{s' \in \mathcal{S}} P(s, s')v(s'), \quad \forall s \in \mathcal{S}_+$$

From above, we split the state summation into state 0 and states $\mathcal{S}_+$:

$$v(s) = R(s) + P(s,0)v(0) + \sum_{s' \in \mathcal{S}_+} P(s,s')v(s')$$

Since $v(0) = 0$ is already established, the term $P(s,0)v(0)$ becomes 0:

$$v(s) = R(s) + \sum_{s' \in \mathcal{S}_+} P(s,s')v(s'), \quad \forall s \in \mathcal{S}_+$$

Now, restricting the system of equations for all non-absorbing states $s \in \mathcal{S}_+$ into vector notation:

$$\hat{v} = \hat{r} + P_2\hat{v}$$

where,

$$\hat{v} : S_+ \to \mathbb{R}^S \quad \text{and} \quad \hat{r} : S_+ \to \mathbb{R}^S, \quad \text{where} \quad \mathcal{S}_+ = S \setminus \{0\}$$

and $P_2 \in \mathbb{R}^{S \times S}$ is the sub-matrix corresponding to transitions between non-absorbing states $\mathcal{S}_+$.

Rearranging the equation and bringing $\hat{v}$ on the left-hand side:

$$\hat{v} - P_2\hat{v} = \hat{r}$$

$$(I - P_2)\hat{v} = \hat{r}$$

Since $(I - P_2)$ is already proven to be non-singular, multiplying both sides by $(I - P_2)^{-1}$ gives:

$$\hat{v} = (I - P_2)^{-1}\hat{r}$$

Combining $v(0) = 0$ and $\hat{v} = (I - P_2)^{-1}\hat{r}$ into the full vector $v \in \mathbb{R}^{S+1}$:

$$v = \begin{bmatrix} v(0) \\ \hat{v} \end{bmatrix} = \begin{bmatrix} 0 \\ (I - P_2)^{-1}\hat{r} \end{bmatrix}$$

3 Solution to Question 3

3.1 (a)

From the previous question, we are given a Markov Reward Process with state space $\mathcal{S} = \{0, 1, \ldots, S\}$, where state 0 is an absorbing state with $R(0) = 0$ and value $v^k(0) = 0$ for all $k \geq 0$.

The value iteration algorithm progresses as:

$$v^{k+1} = r + Pv^k \quad \text{for } k \geq 0$$

where $v^k, r \in \mathbb{R}^{S+1}$.

From the earlier question, the transition matrix $P \in \mathbb{R}^{(S+1)\times(S+1)}$ has the block structure:

$$P = \begin{bmatrix} 1 & \mathbf{0} \\ P_1 & P_2 \end{bmatrix}$$

where $P_1 \in \mathbb{R}^{S\times 1}$ and $P_2 \in \mathbb{R}^{S\times S}$.

The vectors v^k and r are partitioned according to state 0 (the first state) and the non-absorbing states $\mathcal{S}_+ = \{1, \ldots, S\}$ (the remaining S states):

$$v^k = \begin{bmatrix} v^k(0) \\ \hat{v}^k \end{bmatrix} \quad \text{and} \quad r = \begin{bmatrix} r(0) \\ \hat{r} \end{bmatrix}$$

where $\hat{v}^k, \hat{r} \in \mathbb{R}^S$.

Since $r(0) = R(0) = 0$ and $v^0(0) = 0$, we have $v^k(0) = 0$ for all iterations $k \geq 0$.

Substituting the block structures of P, v^k, and r into the value iteration update equation gives:

$$\begin{bmatrix} v^{k+1}(0) \\ \hat{v}^{k+1} \end{bmatrix} = \begin{bmatrix} 0 \\ \hat{r} \end{bmatrix} + \begin{bmatrix} 1 & \mathbf{0} \\ P_1 & P_2 \end{bmatrix} \begin{bmatrix} v^k(0) \\ \hat{v}^k \end{bmatrix}$$

By block matrix-vector multiplication:

$$\begin{bmatrix} 1 & \mathbf{0} \\ P_1 & P_2 \end{bmatrix} \begin{bmatrix} v^k(0) \\ \hat{v}^k \end{bmatrix} = \begin{bmatrix} 1 \cdot v^k(0) + \mathbf{0} \cdot \hat{v}^k \\ P_1 \cdot v^k(0) + P_2 \cdot \hat{v}^k \end{bmatrix}$$

This makes the value iteration update equation in matrix form as:

$$\begin{bmatrix} v^{k+1}(0) \\ \hat{v}^{k+1} \end{bmatrix} = \begin{bmatrix} 0 + 1 \cdot v^k(0) + \mathbf{0} \cdot \hat{v}^k \\ \hat{r} + P_1 \cdot v^k(0) + P_2 \cdot \hat{v}^k \end{bmatrix}$$

$$\implies \begin{bmatrix} v^{k+1}(0) \\ \hat{v}^{k+1} \end{bmatrix} = \begin{bmatrix} 0 + 1 \cdot v^k(0) + \mathbf{0} \cdot \hat{v}^k \\ \hat{r} + P_2 \cdot \hat{v}^k \end{bmatrix}$$

$$\implies \begin{bmatrix} v^{k+1}(0) \\ \hat{v}^{k+1} \end{bmatrix} = \begin{bmatrix} v^k(0) \\ \hat{r} + P_2 \cdot \hat{v}^k \end{bmatrix}$$

In the problem, it is given that $v^0(0) = 0$, by induction $v^k(0) = 0$ holds for all $k \geq 0$. Thus, $v^{k+1}(0) = 0$, confirming state 0 remains unchanged at 0.

For the remaining S rows (Non-Absorbing States $\mathcal{S}_+$), since $P_1 v^k(0) = P_1 \cdot 0 = \mathbf{0}$:

$$\hat{v}^{k+1} = \hat{r} + P_2 \hat{v}^k$$

From the sparsity structure of P from earlier question and the boundary conditions $r(0) = 0$ and $v^0(0) = 0$, the value iteration update on $\mathbb{R}^{S+1}$ reduces equivalently to:

$$\hat{v}^{k+1} = \hat{r} + P_2 \hat{v}^k$$

for all $k \geq 0$, starting from initialization $\hat{v}^0 \in \mathbb{R}^S$.

3.2 (b)

We get $N : \mathcal{S}_+ \to \mathbb{R}$ as the expected number of times state $s \in \mathcal{S}_+$ is visited before reaching the absorbing state 0.

As per the hint, we create an new Markov Reward Process (MRP) with state space $\mathcal{S} = \{0, 1, \ldots, S\}$ and transition matrix P identical to the original process.

New reward function $r_N : \mathcal{S} \to \mathbb{R}$ defined as:

$$r_N(s) = \begin{cases} 1, & \text{if } s \in \mathcal{S}_+ \\ 0, & \text{if } s = 0 \end{cases}$$

Let S_k denote the state at step $k \geq 0$. The total accumulated reward in this new reward process starting from $S_0 = s$ is:

$$\sum_{k=0}^{\infty} r_N(S_k) = \sum_{k=0}^{\infty} \mathbb{I}(S_k \in \mathcal{S}_+)$$

where $\mathbb{I}(\cdot)$ is the indicator function.

The above sum counts 1 for every time step k the process spends in a transient state $\mathcal{S}_+$, and 0 once it reaches the absorbing state 0.

Taking the expected value, the value function of this new MRP is precisely $N(s)$:

$$N(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} r_N(S_k) \;\middle|\; S_0 = s\right]$$

$$N(s) = \mathbb{E}\left[r_N(S_0) + \sum_{k=1}^{\infty} r_N(S_k) \;\middle|\; S_0 = s\right]$$

$$\implies N(s) = r_N(s) + \mathbb{E}\left[\sum_{k=1}^{\infty} r_N(S_k) \;\middle|\; S_0 = s\right]$$

Since $s \in \mathcal{S}_+$, $r_N(s) = 1$.

Applying the law of total expectation by using on the next state $S_1 = s'$:

$$N(s) = 1 + \sum_{s' \in \mathcal{S}} \mathbb{P}(S_1 = s' \mid S_0 = s)\, \mathbb{E}\left[\sum_{k=1}^{\infty} r_N(S_k) \;\middle|\; S_1 = s'\right]$$

$$\implies N(s) = 1 + \sum_{s' \in \mathcal{S}} P(s, s') N(s')$$

We split the summation over all states $\mathcal{S} = \{0\} \cup \mathcal{S}_+$:

$$N(s) = 1 + P(s, 0)N(0) + \sum_{s' \in \mathcal{S}_+} P(s, s') N(s')$$

For the absorbing state 0, $r_N(0) = 0$, $P(0,0) = 1$ and $P(0,s') = 0$ for all $s' \in \mathcal{S}_+$.

Therefore, the value function at state 0 satisfies:

$$N(0) = r_N(0) + P(0,0)N(0) = 0 + 1 \cdot N(0) \implies N(0) = 0$$

Substituting $N(0) = 0$ back, we get:

$$N(s) = 1 + \sum_{s' \in \mathcal{S}_+} P(s,s')N(s') \quad \forall s \in \mathcal{S}_+$$

From earlier problems, we know that the vector of values for transient states in any MRP with reward vector $\hat{r}$ satisfies $\hat{v} = (I - P_2)^{-1}\hat{r}$.

For our new MRP, the reduced reward vector is $\hat{r}_N = \mathbf{1} = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix}^T \in \mathbb{R}^{\mathcal{S}_+}$.

Hence, the vector $\hat{N} = \begin{bmatrix} N(1) & \dots & N(S) \end{bmatrix}^T$ is directly computed as:

$$\hat{N} = (I - P_2)^{-1}\mathbf{1}$$

This is possible to be solved by systems of linear equations.

3.3 (c)

Considering the given mapping $B : \mathbb{R}^S \to \mathbb{R}^S$ is defined by:

$$B(w) = \hat{r} + P_2 w$$

where $\mathcal{S}_+ = \{1, 2, \dots, S\}$ is the set of non-absorbing states.

Since $N(s)$ represents the expected number of visits to state s starting from s, and state s is visited at least once at time $k = 0$, we have $N(s) \geq 1$ for all $s \in \mathcal{S}_+$.

From previous problem, $N(s)$ satisfies:

$$N(s) = 1 + \sum_{s' \in \mathcal{S}_+} P(s,s')N(s')$$

The weighted norm $\|\cdot\|_N$ on $\mathbb{R}^S$ is defined in the problem as:

$$\|w\|_N = \max_{s \in \mathcal{S}_+} \frac{|w(s)|}{N(s)}$$

The contraction parameter α in the problem is defined as:

$$\alpha = \max_{s \in \mathcal{S}_+} \frac{N(s) - 1}{N(s)}$$

Since $N(s) \geq 1$, $N(s) - 1 \geq 0$, which implies:

$$\frac{N(s) - 1}{N(s)} \geq 0 \implies \alpha = \max_{s \in \mathcal{S}_+} \frac{N(s) - 1}{N(s)} \geq 0$$

Because the state space $\mathcal{S}_+$ is finite and $N(s) < \infty$ for all $s \in \mathcal{S}_+$ (since the absorbing state 0 is reachable with probability 1 from every state), $N(s)$ has a maximum finite value:

$$N_{\max} = \max_{s \in \mathcal{S}_+} N(s) < \infty$$

Therefore, for every $s \in \mathcal{S}_+$:

$$\frac{N(s)-1}{N(s)} = 1 - \frac{1}{N(s)} \leq 1 - \frac{1}{N_{\max}} < 1$$

Taking the maximum over all $s \in \mathcal{S}_+$ yields $\alpha < 1$. Hence, $\alpha \in [0, 1)$.
Let $w_1, w_2 \in \mathbb{R}^S$.
By the definition of $B(w)$:

$$B(w_1) - B(w_2) = (\hat{r} + P_2 w_1) - (\hat{r} + P_2 w_2) = P_2(w_1 - w_2)$$

Let $e = w_1 - w_2 \in \mathbb{R}^S$.
By definition of the weighted norm $\|\cdot\|_N$:

$$\|e\|_N = \max_{s' \in \mathcal{S}_+} \frac{|e(s')|}{N(s')} \implies |e(s')| \leq N(s')\|e\|_N \quad \forall s' \in \mathcal{S}_+$$

Now evaluating the norm of $B(w_1) - B(w_2) = P_2 e$:

$$\|B(w_1) - B(w_2)\|_N = \|P_2 e\|_N = \max_{s \in \mathcal{S}_+} \frac{|(P_2 e)(s)|}{N(s)}$$

For any state $s \in \mathcal{S}_+$, expanding the matrix-vector product gives:

$$|(P_2 e)(s)| = \left| \sum_{s' \in \mathcal{S}_+} P(s, s') e(s') \right|$$

Using the triangle inequality and non-negativity of transition probabilities ($P(s, s') \geq 0$):

$$|(P_2 e)(s)| \leq \sum_{s' \in \mathcal{S}_+} P(s, s') |e(s')|$$

Substituting the bound $|e(s')| \leq N(s')\|e\|_N$:

$$|(P_2 e)(s)| \leq \sum_{s' \in \mathcal{S}_+} P(s, s') N(s') \|e\|_N = \left(\sum_{s' \in \mathcal{S}_+} P(s, s') N(s') \right) \|e\|_N$$

From the earlier equation for $N(s)$:

$$\sum_{s' \in \mathcal{S}_+} P(s, s') N(s') = N(s) - 1$$

We can write the inequality equation above as:

$$|(P_2 e)(s)| \leq (N(s) - 1)\|e\|_N$$

Dividing both sides by $N(s)$:

$$\frac{|(P_2 e)(s)|}{N(s)} \leq \frac{N(s) - 1}{N(s)} \|e\|_N$$

Taking the maximum over all $s \in \mathcal{S}_+$:

$$\|B(w_1) - B(w_2)\|_N = \max_{s \in \mathcal{S}_+} \frac{|(P_2 e)(s)|}{N(s)} \leq \left(\max_{s \in \mathcal{S}_+} \frac{N(s) - 1}{N(s)} \right) \|e\|_N = \alpha \|w_1 - w_2\|_N$$

Thus, $B(w)$ is an α-contraction in $\|\cdot\|_N$.

From previous problem, the value iteration sequence is $\hat{v}^{k+1} = B(\hat{v}^k)$.

The true value function $\hat{v}$ is the fixed point satisfying $\hat{v} = B(\hat{v})$, based on previous problems.

Applying the contraction property inductively:

$$\|\hat{v}^{k+1} - \hat{v}\|_N = \|B(\hat{v}^k) - B(\hat{v})\|_N \leq \alpha \|\hat{v}^k - \hat{v}\|_N$$

By induction over k:

$$\|\hat{v}^k - \hat{v}\|_N \leq \alpha^k \|\hat{v}^0 - \hat{v}\|_N$$

Since $v^k = \begin{bmatrix} 0 \\ \hat{v}^k \end{bmatrix}$ and $v = \begin{bmatrix} 0 \\ \hat{v} \end{bmatrix}$, the full value vector has the linear rate of convergence:

$$\|v^k - v\|_N \leq \alpha^k \|v^0 - v\|_N$$

3.4 (d)

From previous question, the contraction factor is given by:

$$\alpha = \max_{s \in \mathcal{S}_+} \frac{N(s) - 1}{N(s)} = 1 - \frac{1}{\max_{s \in \mathcal{S}_+} N(s)} = 1 - \frac{1}{N_{\max}}$$

where $N_{\max} = \max_{s \in \mathcal{S}_+} N(s) \geq 1$.

Also, the error after k iterations satisfies the linear convergence bound:

$$\|v^k - v\|_N \leq \alpha^k \|v^0 - v\|_N$$

To guarantee an ϵ-accurate solution, we require:

$$\alpha^k \|v^0 - v\|_N \leq \epsilon$$

Rearranging for α^k:

$$\alpha^k \leq \frac{\epsilon}{\|v^0 - v\|_N}$$

Taking the (log) on both sides:

$$k \log(\alpha) \leq \log\left(\frac{\epsilon}{\|v^0 - v\|_N}\right) = -\log\left(\frac{\|v^0 - v\|_N}{\epsilon}\right)$$

Since $\alpha \in [0, 1)$, $\log(\alpha) < 0$.

Dividing both sides by $\log(\alpha)$ flips the inequality sign:

$$k \geq \frac{\log\left(\frac{\|v^0 - v\|_N}{\epsilon}\right)}{-\log(\alpha)} = \frac{\log\left(\frac{\|v^0 - v\|_N}{\epsilon}\right)}{\log(1/\alpha)}$$

By checking the first derivative of the function:

$$\log(1 - x) \leq -x \quad \text{for all } x < 1$$

Substituting $x = \frac{1}{N_{\max}}$ from previous question gives:

$$\log(\alpha) = \log\left(1 - \frac{1}{N_{\max}}\right) \leq -\frac{1}{N_{\max}}$$

Multiplying both sides by -1 yields:

$$-\log(\alpha) = \log\left(\frac{1}{\alpha}\right) \geq \frac{1}{N_{\max}}$$

Taking the reciprocal on both sides gives:

$$\frac{1}{-\log(\alpha)} \leq N_{\max}$$

Substituting this lower bound back into our inequality for k:

$$k \geq N_{\max} \ln\left(\frac{\|v^0 - v\|_N}{\epsilon}\right)$$

Substituting this lower bound into our inequality for k:

$$k \geq N_{\max} \log\left(\frac{\|v^0 - v\|_N}{\epsilon}\right)$$

Using an ϵ-accurate solution $\|v^k - v\|_N \leq \epsilon$ is guaranteed after:

$$k = N_{\max} \log\left(\frac{\|v^0 - v\|_N}{\epsilon}\right) \text{ iterations}$$

Hence, by strictly bounding the contraction factor $\alpha = 1 - 1/N_{\max}$ via the standard inequality $\ln(1 - x) \leq -x$ (through first order derivative test), the required number of iterations to achieve $\|v^k - v\|_N \leq \epsilon$ is $k = N_{\max} \log\left(\frac{\|v^0 - v\|_N}{\epsilon}\right)$.

4 Solution to Question 4

We consider a MRP with states $\mathcal{S} = \{1, \ldots, S\}$, transition matrix $P \in \mathbb{R}^{S \times S}$, reward function $R : \mathcal{S} \to \mathbb{R}$.

We also consider some discount factor $\gamma \in [0, 1)$.

From the literature and notes, the value function for a given P and R is:

$$v = (I - \gamma P)^{-1} R$$

4.1 (a)

Suppose we know P as true transition matrix, but only have an approximation $\hat{R}$ of the reward function.

It satisfies $\|R - \hat{R}\|_\infty \leq \epsilon$.

We compute the estimated value as:

$$\hat{v} = (I - \gamma P)^{-1} \hat{R}$$

We need to prove:

$$\|\hat{v} - v\|_\infty \leq \frac{\epsilon}{1 - \gamma}$$

By the linearity of matrix-vector multiplication, the error vector can be written as:

$$\hat{v} - v = (I - \gamma P)^{-1} \hat{R} - (I - \gamma P)^{-1} R = (I - \gamma P)^{-1} (\hat{R} - R)$$

Taking the vector infinity norm $\|\cdot\|_\infty$ on both sides and applying the property $\|Ax\| \leq \|A\|\|x\|$ for induced matrix norms (given in the Note):

$$\|\hat{v} - v\|_\infty = \|(I - \gamma P)^{-1} (\hat{R} - R)\|_\infty \leq \|(I - \gamma P)^{-1}\|_\infty \|\hat{R} - R\|_\infty$$

Since $\|R - \hat{R}\|_\infty = \|\hat{R} - R\|_\infty \leq \epsilon$, we have:

$$\|\hat{v} - v\|_\infty \leq \|(I - \gamma P)^{-1}\|_\infty \cdot \epsilon$$

Let $B \in \mathbb{R}^{S \times S}$ be a matrix with $\|B\| < 1$.

We start from the identity provided in the Hint:

$$(I + B)^{-1} + B(I + B)^{-1} = I$$

Isolating $(I + B)^{-1}$ on the left-hand side gives:

$$(I + B)^{-1} = I - B(I + B)^{-1}$$

Taking matrix norm $\|\cdot\|$ on both sides:

$$\|(I + B)^{-1}\| = \|I - B(I + B)^{-1}\|$$

Applying the triangle inequality $\|X - Y\| \leq \|X\| + \|Y\|$:

$$\|(I + B)^{-1}\| \leq \|I\| + \|B(I + B)^{-1}\|$$

By the sub-multiplicative property of induced matrix norms $\|XY\| \leq \|X\|\|Y\|$ and the fact that $\|I\| = 1$ for induced I:

$$\|(I+B)^{-1}\| \leq 1 + \|B\|\|(I+B)^{-1}\|$$

Subtracting $\|B\|\|(I+B)^{-1}\|$ from both sides:

$$\|(I+B)^{-1}\| - \|B\|\|(I+B)^{-1}\| \leq 1$$

$$(1 - \|B\|)\|(I+B)^{-1}\| \leq 1$$

Since $\|B\| < 1$, we have $1 - \|B\| > 0$.
Dividing both sides by $(1 - \|B\|)$ gives:

$$\|(I+B)^{-1}\| \leq \frac{1}{1-\|B\|}$$

Let $B = -\gamma P$.
We then calculate $\|B\|_\infty$:

$$\|B\|_\infty = \| - \gamma P\|_\infty = \gamma \|P\|_\infty$$

Using the definition of the induced matrix infinity norm given in the Note:

$$\|P\|_\infty = \max_{i=1,\ldots,S} \sum_{j=1}^{S} |P_{ij}|$$

Since P is a transition probability matrix (row-stochastic), all elements are non-negative ($P_{ij} \geq 0$) and each row sums to 1 ($\sum_{j=1}^{S} P_{ij} = 1$ for all i).
Therefore:

$$\|P\|_\infty = \max_{i=1,\ldots,S}(1) = 1$$

Given $\gamma \in [0, 1)$, this implies:

$$\|B\|_\infty = \gamma(1) = \gamma < 1$$

Now, applying the bound with $B = -\gamma P$:

$$\|(I - \gamma P)^{-1}\|_\infty \leq \frac{1}{1 - \| - \gamma P\|_\infty} = \frac{1}{1-\gamma}$$

Substituting $\|(I - \gamma P)^{-1}\|_\infty \leq \frac{1}{1-\gamma}$ back into value equation:

$$\|\hat{v} - v\|_\infty \leq \frac{1}{1-\gamma} \cdot \epsilon = \frac{\epsilon}{1-\gamma}$$

4.2 (b)

Suppose that we have an approximation $\hat{v}$ of the true value function v. Suppose that $\|\hat{v} - B(\hat{v})\|_\infty \leq \epsilon$, where $B(v) = R + \gamma P v$ is the Bellman operator.

We have to prove that:

$$\|\hat{v} - v\|_\infty \leq \frac{\epsilon}{1-\gamma}$$

By definition, the true value function v satisfies the Bellman equation:

$$v = R + \gamma P v$$

Therefore, v is a fixed point of the Bellman operator B, meaning:

$$B(v) = v$$

Using the above property, we can write the vector difference $\hat{v} - v$ as:

$$\hat{v} - v = \hat{v} - B(\hat{v}) + B(\hat{v}) - v$$

$$\hat{v} - v = (\hat{v} - B(\hat{v})) + (B(\hat{v}) - B(v))$$

Using the given definition $B(v) = R + \gamma P v$, we compute the difference:

$$B(\hat{v}) - B(v) = (R + \gamma P \hat{v}) - (R + \gamma P v) = \gamma P(\hat{v} - v)$$

Substituting this back into the expression for $\hat{v} - v$ gives:

$$\hat{v} - v = (\hat{v} - B(\hat{v})) + \gamma P(\hat{v} - v)$$

Rearranging the terms containing $(\hat{v} - v)$ to the left-hand side:

$$(I - \gamma P)(\hat{v} - v) = \hat{v} - B(\hat{v})$$

Since P is a valid transition matrix (with non-negative entries and row sums equal to 1) and $\gamma \in [0, 1)$, the matrix $(I - \gamma P)$ is invertible (also proven earlier).

Multiplying both sides by $(I - \gamma P)^{-1}$, we obtain:

$$\hat{v} - v = (I - \gamma P)^{-1}(\hat{v} - B(\hat{v}))$$

Taking the infinity norm $\|\cdot\|_\infty$ on both sides and applying the sub-multiplicative property of induced matrix norms given earlier ($\|Ax\| \leq \|A\|\|x\|$):

$$\|\hat{v} - v\|_\infty = \|(I - \gamma P)^{-1}(\hat{v} - B(\hat{v}))\|_\infty \leq \|(I - \gamma P)^{-1}\|_\infty \|\hat{v} - B(\hat{v})\|_\infty$$

As given:

$$\|\hat{v} - B(\hat{v})\|_\infty \leq \epsilon$$

Thus:

$$\|\hat{v} - v\|_\infty \leq \|(I - \gamma P)^{-1}\|_\infty \cdot \epsilon$$

From previous solution, we have $B = -\gamma P$.
We also have $\|B\|_\infty < 1$ and $\|(I+B)^{-1}\|_\infty \leq \frac{1}{1-\|B\|_\infty}$.
We also saw:

$$\|B\|_\infty = \| - \gamma P\|_\infty = \gamma \|P\|_\infty$$

Applying same logic, since P is a row-stochastic matrix (transition probability matrix), the sum of entries in each row is $\sum_{j=1}^{S} P_{ij} = 1$.
Thus, by the definition of the matrix ∞-norm:

$$\|P\|_\infty = \max_{i=1,\dots,S} \sum_{j=1}^{S} |P_{ij}| = 1$$

Therefore:

$$\|B\|_\infty = \gamma \cdot 1 = \gamma$$

Since $\gamma \in [0, 1)$, we have $\|B\|_\infty < 1$, which satisfies the condition for the matrix bound from above:

$$\|(I - \gamma P)^{-1}\|_\infty \leq \frac{1}{1-\gamma}$$

Substituting the matrix norm bound back into our inequality:

$$\|\hat{v} - v\|_\infty \leq \frac{1}{1-\gamma} \cdot \epsilon = \frac{\epsilon}{1-\gamma}$$

4.3 (c)

Let $v = (I - \gamma P)^{-1} R$ be the true discounted value and given $\hat{v} = (I - \gamma \hat{P})^{-1} R$ as the approximated value, where $P, \hat{P} \in \mathbb{R}^{S \times S}$ are row-stochastic transition matrices with $\hat{P}$ as the approximate transition matrix, $R \in \mathbb{R}^S$ is the true reward, and $\gamma \in [0, 1)$ is the discount factor.
Given that $\|P - \hat{P}\|_\infty \leq \epsilon$, we have to prove:

$$\|\hat{v} - v\|_\infty \leq \frac{\gamma\epsilon}{1-\gamma}$$

From previous solutions, we already have:

$$\|(I - \gamma P)^{-1}\|_\infty \leq \frac{1}{1-\gamma}$$

This can be written for $\hat{P}$ as:

$$\|(I - \gamma \hat{P})^{-1}\|_\infty \leq \frac{1}{1-\gamma}$$

By the given definition of the value functions:

$$(I - \gamma \hat{P})\hat{v} = R \quad \text{and} \quad (I - \gamma P)v = R$$

Equating both expressions for R:

$$(I - \gamma\hat{P})\hat{v} = (I - \gamma P)v$$

Expanding both sides:

$$\hat{v} - \gamma\hat{P}\hat{v} = v - \gamma Pv$$

Rearranging to isolate $\hat{v} - v$ on the left side:

$$\hat{v} - v = \gamma\hat{P}\hat{v} - \gamma Pv$$

Adding and subtracting $\gamma\hat{P}v$ on the right side:

$$\hat{v} - v = \gamma\hat{P}\hat{v} - \gamma\hat{P}v + \gamma\hat{P}v - \gamma Pv$$

$$\hat{v} - v = \gamma\hat{P}(\hat{v} - v) + \gamma(\hat{P} - P)v$$

Grouping the $(\hat{v} - v)$ terms:

$$(I - \gamma\hat{P})(\hat{v} - v) = \gamma(\hat{P} - P)v$$

We already know that $(I - \gamma\hat{P})$ is invertible.
Multiplying both sides by $(I - \gamma\hat{P})^{-1}$:

$$\hat{v} - v = \gamma(I - \gamma\hat{P})^{-1}(\hat{P} - P)v$$

Taking the vector ∞-norm on both sides and using the compatibility property $\|Ax\|_\infty \leq \|A\|_\infty\|x\|_\infty$:

$$\|\hat{v} - v\|_\infty = \left\|\gamma(I - \gamma\hat{P})^{-1}(\hat{P} - P)v\right\|_\infty$$

$$\|\hat{v} - v\|_\infty \leq \gamma \left\|(I - \gamma\hat{P})^{-1}\right\|_\infty \|\hat{P} - P\|_\infty\|v\|_\infty$$

From the previous solution, $\left\|(I - \gamma\hat{P})^{-1}\right\|_\infty \leq \frac{1}{1-\gamma}$.
From the problem, $\|P - \hat{P}\|_\infty = \|\hat{P} - P\|_\infty \leq \epsilon$.
Combining these bounds:

$$\|\hat{v} - v\|_\infty \leq \gamma \cdot \left(\frac{1}{1-\gamma}\right) \cdot \epsilon \cdot \|v\|_\infty$$

When we have $\|v\|_\infty \leq 1$, we get:

$$\|\hat{v} - v\|_\infty \leq \frac{\gamma\epsilon}{1-\gamma}$$

5 Solution to Question 5

We compare the computational efficiency of Dynamic Programming (Value Iteration) versus Solving a Linear System (like Gaussian Elimination) for both discounted Markov Reward Processes (MRPs) and shortest-path MRPs under dense matrix assumptions ($S \times S$ dense transition matrices).

We analyze the time and space complexity of both approaches for a state space of size S.

When **solving the linear system of equations**:

From previous questions, the value vector is computed by solving:

$$(I - \gamma P)v = R \quad \text{or} \quad (I - P_2)\hat{v} = \hat{r}$$

Time Complexity: For a dense $S \times S$ matrix, solving the linear system via standard Gaussian elimination (LU decomposition) or explicit matrix inversion takes $O(S^3)$ time.

Space Complexity: $O(S^2)$ to store the dense transition and identity matrices.

Key Characteristic: The cost is independent of the discount factor γ or the required precision ϵ, and it gives an exact solution in a fixed polynomial number of operations.

When **solving using dynamic programming**:

From previous questions and notes, the value iteration algorithm updates estimates via:

$$v^{k+1} = R + \gamma P v^k \quad \text{or} \quad \hat{v}^{k+1} = \hat{r} + P_2 \hat{v}^k$$

Cost per Iteration: Each iteration involves a dense matrix-vector multiplication, which takes $O(S^2)$ time.

Number of Iterations (k): To achieve an ϵ-accurate solution ($||v^k - v|| \leq \epsilon$):

Discounted MRP: From the contraction rate $\gamma \in [0, 1)$, the number of iterations required is $k = O\left(\frac{1}{1-\gamma} \log\left(\frac{1}{\epsilon}\right)\right)$.

Shortest-path MRP: The number of iterations required is $k = O\left(N_{\max} \log\left(\frac{1}{\epsilon}\right)\right)$, where $N_{\max} = \max_{s \in \mathcal{S}_+} N(s)$ is the maximum expected number of visits to any transient state before reaching absorption.

Total Time Complexity:

Discounted MRP: $O\left(S^2 \cdot \frac{1}{1-\gamma} \log\left(\frac{1}{\epsilon}\right)\right)$

Shortest-path MRP: $O\left(S^2 \cdot N_{\max} \log\left(\frac{1}{\epsilon}\right)\right)$

Space Complexity: $O(S)$ to store the value vectors and $O(S^2)$ to store the transition matrix. So it needs $O(S^2)$.

Efficiency comparison and conditions

For both the methods, the space complexity is dominated by $O(S^2)$, hence both them are equivalent in terms of storage.

For time complexity, now we will compare the $O(S^3)$ linear system cost against the $O\left(S^2 \cdot k\right)$ dynamic programming cost. Since $O(S^2)$ is common in both, we have to compare only $O(S)$ vs. $O(k)$.

Dynamic programming is efficient when $k \ll S$

If large ϵ is acceptable (higher approximation error), then $\log\left(\frac{1}{\epsilon}\right)$ decreases.

For discounted MRP, this means $\frac{1}{1-\gamma} \ll S$, and for shortest-path MRP, this means $N_{\max} \ll S$.

Solving linear equations is efficient when $k \gg S$

If small ϵ is desirable (low approximation error), then $\log\left(\frac{1}{\epsilon}\right)$ increases.

For discounted MRP, this means $\frac{1}{1-\gamma} \gg S$, and for shortest-path MRP, this means $N_{\max} \gg S$.

Solving linear equations and dynamic programming is similar when $k \approx S$

When $\frac{1}{1-\gamma} = \Theta(S)$ for discounted MRP, and $N_{\max} = \Theta(S)$ for shortest-path MRP, which seems possible when the $|S|$ is comparable to k.

6 Solution to Question 6

Given the optimal Bellman operator for a discounted Markov Decision Process (MDP):

$$\mathcal{T}(Q)(s,a) = r(s,a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} \left(\max_{a' \in \mathcal{A}} Q(s',a') \right)$$

where $\gamma \in [0,1)$.

We have to prove that $\mathcal{T}(\cdot)$ is a contraction in the $||\cdot||_\infty$ with parameter γ and has a unique fixed point Q^*, representing the optimal Q-table.

Let $\mathcal{Q} = \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ be the vector space of all bounded Q-tables $Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$.

With infinity norm, defined as:

$$||Q||_\infty = \max_{s \in \mathcal{S},\, a \in \mathcal{A}} |Q(s,a)|$$

The space $(\mathcal{Q}, ||\cdot||_\infty)$ is a Banach space (complete normed vector space).

To prove the contraction of $\mathcal{T}$, we use the standard property of the maximum operator for contraction for any two bounded functions $f_1, f_2 : \mathcal{A} \to \mathbb{R}$:

$$\left| \max_{a' \in \mathcal{A}} f_1(a') - \max_{a' \in \mathcal{A}} f_2(a') \right| \leq \max_{a' \in \mathcal{A}} |f_1(a') - f_2(a')|$$

Let $Q_1, Q_2 \in \mathcal{Q}$ be any two arbitrary Q-tables.

For any state-action pair $(s,a) \in \mathcal{S} \times \mathcal{A}$, we evaluate the absolute difference under the optimal Bellman operator $\mathcal{T}$:

$$|\mathcal{T}(Q_1)(s,a) - \mathcal{T}(Q_2)(s,a)| =$$

$$\left| r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s,a) \max_{a' \in \mathcal{A}} Q_1(s',a') - \left(r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s,a) \max_{a' \in \mathcal{A}} Q_2(s',a') \right) \right|$$

Canceling the reward term $r(s,a)$ and factoring out γ:

$$= \gamma \left| \sum_{s' \in \mathcal{S}} P(s'|s,a) \left(\max_{a' \in \mathcal{A}} Q_1(s',a') - \max_{a' \in \mathcal{A}} Q_2(s',a') \right) \right|$$

Using the triangle inequality for sums:

$$\leq \gamma \sum_{s' \in \mathcal{S}} P(s'|s,a) \left| \max_{a' \in \mathcal{A}} Q_1(s',a') - \max_{a' \in \mathcal{A}} Q_2(s',a') \right|$$

Applying the max-norm bound from previous step:

$$\left| \max_{a' \in \mathcal{A}} Q_1(s',a') - \max_{a' \in \mathcal{A}} Q_2(s',a') \right| \leq \max_{a'' \in \mathcal{A}} |Q_1(s',a'') - Q_2(s',a'')| \leq ||Q_1 - Q_2||_\infty$$

Substituting this upper bound back into the summation:

$$|\mathcal{T}(Q_1)(s,a) - \mathcal{T}(Q_2)(s,a)| \leq \gamma \sum_{s' \in \mathcal{S}} P(s'|s,a) ||Q_1 - Q_2||_\infty$$

Because this inequality holds for all $(s,a) \in \mathcal{S} \times \mathcal{A}$, we can take the maximum over all state-action pairs on the left-hand side:

$$||\mathcal{T}(Q_1) - \mathcal{T}(Q_2)||_\infty = \max_{s,a} |\mathcal{T}(Q_1)(s,a) - \mathcal{T}(Q_2)(s,a)| \leq \gamma ||Q_1 - Q_2||_\infty$$

Since $\gamma \in [0,1)$, $\mathcal{T}$ is a γ-contraction mapping in the infinity norm.

By the Banach Fixed-Point Theorem, $\mathcal{T}$ possesses a unique fixed point $Q^* \in \mathcal{Q}$ such that $\mathcal{T}(Q^*) = Q^*$, which corresponds to the optimal Q-table.

By using the Q-value iteration sequence by $Q^{k+1} = \mathcal{T}(Q^k)$ for $k = 0, 1, 2, \ldots$, starting from an arbitrary initial table $Q^0 \in \mathcal{Q}$.

We want to bound the distance between the kth iteration Q^k and the optimal fixed point Q^*:

$$||Q^k - Q^*||_\infty = ||\mathcal{T}(Q^{k-1}) - \mathcal{T}(Q^*)||_\infty$$

Applying the contraction property proven from above:

$$||Q^k - Q^*||_\infty \leq \gamma ||Q^{k-1} - Q^*||_\infty$$

By recursively applying this inequality k times down to the initial estimate Q^0:

$$||Q^k - Q^*||_\infty \leq \gamma^k ||Q^0 - Q^*||_\infty$$

Since $\gamma \in [0,1)$, $\lim_{k \to \infty} \gamma^k = 0$, proving that value iteration Q^k converges linearly to the optimal Q-table Q^* with a contraction factor of γ^k.

7 Solution to Question 7

In the question, we are given a Markov Reward Process (MRP) with state space $\mathcal{S} = \{1, 2\}$ and reward vector $r = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, meaning $r(1) = 1$ and $r(2) = -1$. Transition probability matrix is given as

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

This represents a deterministic, periodic Markov chain of period 2, where state $P(1,2) = 1$, $P(2,1) = 1$, $P(1,1) = 0$ and $P(2,2) = 0$.

7.1 (a)

As given in the question, the gain $g(s)$ represents the long-term average reward starting from state $S_1 = s$:

$$g(s) = \lim_{K\to\infty} \frac{1}{K}\mathbb{E}\left[\sum_{k=1}^{K} r(S_k)\Big|S_1 = s\right]$$

If $S_1 = 1$, then in general, the sequence of rewards will become deterministically: $1, -1, 1, -1, \ldots$.

The cumulative sum $\sum_{k=1}^{K} r(S_k)$ is bounded between 0 and 1 for all K:

$$\sum_{k=1}^{K} r(S_k) \in \{0, 1\}$$

Therefore, the gain for $s = 1$ is:

$$g(1) = \lim_{K\to\infty} \frac{1}{K}\mathbb{E}\left[\sum_{k=1}^{K} r(S_k)\Big|S_1 = 1\right] = \lim_{K\to\infty} \frac{\text{bounded value}}{K} = 0$$

For $S_1 = 2$, the rewards will become: $-1, 1, -1, 1, \ldots$, and the cumulative sum $\sum_{k=1}^{K} r(S_k)$ is bounded between -1 and 0:

$$\sum_{k=1}^{K} r(S_k) \in \{-1, 0\}$$

Therefore, the gain for $s = 2$ is:

$$g(2) = \lim_{K\to\infty} \frac{1}{K}\mathbb{E}\left[\sum_{k=1}^{K} r(S_k)\Big|S_1 = 2\right] = \lim_{K\to\infty} \frac{\text{bounded value}}{K} = 0$$

Thus, the gain vector is $g = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Since the Markov chain is periodic, the standard limit of the cumulative reward differences does not converge in the ordinary sense.

Instead, the bias $h(s)$ is defined using the given Cesaro limit (C-lim):

$$h(s) = \text{C-lim}_{K\to\infty}\mathbb{E}\left[\sum_{k=1}^{K} \left(r(S_k) - g(S_k)\right)\Big|S_1 = s\right]$$

Since $g(s) = 0$ for all states, this simplifies to the Cesaro limit of the cumulative reward sums $y_K(s) = \sum_{k=1}^{K} r(S_k)$ given starting state s:

$$h(s) = \text{C-lim}_{K\to\infty} y_K(s) = \lim_{N\to\infty} \frac{1}{N} \sum_{K=1}^{N} y_K(s)$$

If $S_1 = 1$, the partial sums $y_K(1)$ are $1, 0, 1, 0, \ldots$.
In general, $y_{2m} = 0$ and $y_{2m+1} = 1$.
To compute the Cesaro limit as $N \to \infty$, we can evaluate the average of the first N partial sums.
When $N = 2m$ (even):

$$\frac{1}{2m} \sum_{K=1}^{2m} y_K(1) = \frac{m \times 1 + m \times 0}{2m} = \frac{m}{2m} = \frac{1}{2}$$

When $N = 2m + 1$ (odd):

$$\frac{1}{2m+1} \sum_{K=1}^{2m+1} y_K(1) = \frac{m+1}{2m+1} \xrightarrow{m \to \infty} \frac{1}{2}$$

Taking the limit as $N \to \infty$, we get:

$$h(1) = \frac{1}{2} = 0.5$$

If $S_1 = 2$, the partial sums $y_K(1)$ are $-1, 0, -1, 0, \ldots$.
In general, $y_{2m} = 0$ and $y_{2m+1} = -1$.
We evaluate the Cesaro average
When $N = 2m$ (even):

$$\frac{1}{2m} \sum_{K=1}^{2m} y_K(2) = \frac{m \times (-1) + m \times 0}{2m} = \frac{-m}{2m} = -\frac{1}{2}$$

When $N = 2m + 1$ (odd):

$$\frac{1}{2m+1} \sum_{K=1}^{2m+1} y_K(2) = \frac{-m-1}{2m+1} \xrightarrow{m \to \infty} -\frac{1}{2}$$

Taking the limit as $N \to \infty$, we get:

$$h(2) = -\frac{1}{2} = -0.5$$

Thus, the bias vector is $h = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$.

7.2 (b)

A probability vector $\pi = \begin{bmatrix} \pi(1) & \pi(2) \end{bmatrix}^T$ is a stationary distribution for a Markov chain with transition matrix P if and only if it satisfies two fundamental conditions:
(1) The distribution remains invariant under one-step transitions:

$$\pi^T P = \pi^T$$

.

(2) The components sum to 1,

$$\sum_{s\in\mathcal{S}} \pi(s) = \pi(1) + \pi(2) = 1 \quad \text{and} \quad \pi(s) \geq 0 \text{ for all } s \in \mathcal{S}$$

Expanding the matrix equation $\pi^T P = \pi^T$:

$$\begin{bmatrix}\pi(1) & \pi(2)\end{bmatrix} \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}\pi(1) & \pi(2)\end{bmatrix}$$

This gives the following system of linear equations by matching components, $\pi(2) = \pi(1)$ and $\pi(1) = \pi(2)$.

Both equations simplify to the single constraint:

$$\pi(1) = \pi(2)$$

Now, substituting this relation into the summation condition ($\pi(1) + \pi(2) = 1$):

$$\pi(1) + \pi(1) = 1 \implies 2\pi(1) = 1 \implies \pi(1) = \frac{1}{2}$$

Similarly, we find:

$$\pi(2) = \frac{1}{2}$$

Thus, the unique stationary distribution of the Markov chain is:

$$\pi = \begin{bmatrix}\frac{1}{2}\\ \frac{1}{2}\end{bmatrix}$$

For a finite, irreducible MRP, the long-term average reward (or gain g) is constant across all states and is given by the expected reward under the stationary distribution:

$$g = \sum_{s\in\mathcal{S}} \pi(s) r(s) = \pi^T r$$

Substituting the stationary distribution π and the reward vector r:

$$g = \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\end{bmatrix} \begin{bmatrix}1\\ -1\end{bmatrix}$$

Evaluating the product:

$$g = \left(\frac{1}{2} \times 1\right) + \left(\frac{1}{2} \times (-1)\right) = \frac{1}{2} - \frac{1}{2} = 0$$

So we have for a stationary distribution:
$\pi = \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\end{bmatrix}^T$
Gain $g = \mathbf{0}$

7.3 (c)

We need to compute the discounted rewards $v_\lambda(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} \lambda^k r(S_k) | S_0 = s\right]$ for $\lambda \in [0, 1)$ and use them to find the gain $g(s)$.

We assume that the discounted reward vector $v_\lambda \in \mathbb{R}^2$ satisfies the standard Bellman equation for discounted MRPs:

$$v_\lambda = r + \lambda P v_\lambda$$

Rearranging the terms gives the linear system:

$$(I - \lambda P) v_\lambda = r$$

Substituting the given transition matrix P and reward vector r:

$$\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right) \begin{bmatrix} v_\lambda(1) \\ v_\lambda(2) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\lambda \\ -\lambda & 1 \end{bmatrix} \begin{bmatrix} v_\lambda(1) \\ v_\lambda(2) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$v_\lambda(1) = 1 + \lambda v_\lambda(2)$$

$$-\lambda(1 + \lambda v_\lambda(2)) + v_\lambda(2) = -1$$

$$-\lambda - \lambda^2 v_\lambda(2) + v_\lambda(2) = -1$$

$$v_\lambda(2)(1 - \lambda^2) = \lambda - 1$$

Since $\lambda \in [0, 1)$, we know that $1 - \lambda^2 = (1 - \lambda)(1 + \lambda) \neq 0$.
Dividing both sides by $1 - \lambda^2$:

$$v_\lambda(2) = \frac{\lambda - 1}{(1 - \lambda)(1 + \lambda)} = \frac{-(1 - \lambda)}{(1 - \lambda)(1 + \lambda)} = -\frac{1}{1 + \lambda}$$

Now, substituting $v_\lambda(2)$ back into the expression for $v_\lambda(1)$:

$$v_\lambda(1) = 1 + \lambda \left(-\frac{1}{1 + \lambda} \right) = 1 - \frac{\lambda}{1 + \lambda} = \frac{1 + \lambda - \lambda}{1 + \lambda} = \frac{1}{1 + \lambda}$$

Thus, the discounted reward vector for any $\lambda \in [0, 1)$ is:

$$v_\lambda = \begin{bmatrix} \frac{1}{1+\lambda} \\ -\frac{1}{1+\lambda} \end{bmatrix}$$

Using the Laurent series expansion for discounted MRP, we get:

$$v_\lambda(s) = \frac{1}{1 - \lambda} g(s) + h(s) + \mathcal{O}(1 - \lambda)$$

Because the leading term of $v_\lambda(s)$ is scaled by $\frac{1}{1-\lambda}$, we can isolate the gain $g(s)$ by multiplying the discounted value by $(1-\lambda)$ and taking the limit as λ approaches 1:

$$g(s) = \lim_{\lambda \uparrow 1}(1-\lambda)v_\lambda(s)$$

This helps to bridge the gap between discounted evaluation and long-term average reward (gain).

Now, if we use the limit formula for each state,
for state 1 ($s = 1$):

$$g(1) = \lim_{\lambda \uparrow 1}(1-\lambda)\left(\frac{1}{1+\lambda}\right) = \frac{1-1}{1+1} = \frac{0}{2} = 0$$

for state 2 ($s = 2$):

$$g(2) = \lim_{\lambda \uparrow 1}(1-\lambda)\left(-\frac{1}{1+\lambda}\right) = -\frac{1-1}{1+1} = 0$$

Thus, the long-term average gain for both states is 0.

7.4 (d)

We are given the linear system for the gain g and bias vector $h = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$:

$$g\mathbf{1} + (I - P)h = r$$

This becomes a matrix equation $Ax = b$, where $x = \begin{bmatrix} g \\ h_1 \\ h_2 \end{bmatrix}$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} g \\ h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

From the theory of linear systems, if x_0 is any particular solution to $Ax = b$, then the complete set of solutions x can be expressed as:

$$x = x_0 + z$$

where $z \in \mathcal{N}(A)$ represents any vector in the null space (kernel) of A, meaning $Az = 0$.

To find $\mathcal{N}(A)$, we solve the homogeneous system $Az = 0$ for $z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the following system of scalar equations: $z_1 + z_2 - z_3 = 0$
$z_1 - z_2 + z_3 = 0$
Adding both the equations together eliminates z_2 and z_3:

$$(z_1 + z_2 - z_3) + (z_1 - z_2 + z_3) = 0 \implies 2z_1 = 0 \implies z_1 = 0$$

Substituting $z_1 = 0$ back into equation:

$$0 + z_2 - z_3 = 0 \implies z_2 = z_3$$

Thus, choosing $z_2 = z_3 = c$ for any arbitrary scalar $c \in \mathbb{R}$, the null space is given by:

$$\mathcal{N}(A) = \left\{ c \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \mid c \in \mathbb{R} \right\}$$

Let $x = \begin{bmatrix} g \\ h_1 \\ h_2 \end{bmatrix}$ be an arbitrary solution to $Ax = b$, and let $x_0 = \begin{bmatrix} g_0 \\ h_{1,0} \\ h_{2,0} \end{bmatrix}$ be a particular solution.

Using our null space structure:

$$\begin{bmatrix} g \\ h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} g_0 \\ h_{1,0} \\ h_{2,0} \end{bmatrix} + \begin{bmatrix} 0 \\ c \\ c \end{bmatrix} = \begin{bmatrix} g_0 \\ h_{1,0} + c \\ h_{2,0} + c \end{bmatrix}$$

By looking at the first component corresponding to g:

$$g = g_0 + 0 = g_0$$

Because the first element of any null space vector z is strictly 0 from above, the value of g remains $g = g_0$ across every possible solution to the linear system.

To find the numerical value of g, we see the two scalar equations represented by the matrix system:
$g + h_1 - h_2 = 1$
$g - h_1 + h_2 = -1$
Adding and solving for both the equations gives us $g = 0$.

Hence, through null space, all solutions have the same value of g because the first element z_1 of the null space matrix is 0, and $g = 0$ across solutions.

8 Solution to Question 8

Our given condition is an average reward Markov reward process $(\mathcal{S}, P, r)$ with state space $\mathcal{S}$.

The span semi-norm of a vector $v \in \mathbb{R}^{\mathcal{S}}$ is defined as:

$$\text{sp}(v) = \max_{s \in \mathcal{S}} v(s) - \min_{s \in \mathcal{S}} v(s)$$

Assuming that P has the property that there exists $\gamma \in [0, 1)$ such that:

$$\text{sp}(Pv) \leq \gamma \, \text{sp}(v)$$

for all vectors v.

By the definition of the target gain g^* and bias h^* given in the question, they satisfy the average-reward equation:

$$g^*\mathbf{1} + h^* = r + Ph^*$$

8.1 (a)

We are given the Value Iteration (VI) update rule starting from an initial bias estimate h^0:

$$h^{k+1} = L(h^k) := r + Ph^k$$

The error vector becomes $h^{k+1} - h^*$.

From the Value Iteration update rule, we have:

$$h^{k+1} = r + Ph^k$$

From the above equation, we are given:

$$g^*\mathbf{1} + h^* = r + Ph^*$$

Rearranging this expression to isolate h^*:

$$h^* = r + Ph^* - g^*\mathbf{1}$$

Now, substituting both expressions into the error term $h^{k+1} - h^*$:

$$h^{k+1} - h^* = (r + Ph^k) - (r + Ph^* - g^*\mathbf{1})$$

Simplifying the right-hand side by canceling r:

$$h^{k+1} - h^* = P(h^k - h^*) + g^*\mathbf{1}$$

Taking the span semi-norm $\text{sp}(\cdot)$ on both sides of the error equation:

$$\text{sp}(h^{k+1} - h^*) = \text{sp}\left(P(h^k - h^*) + g^*\mathbf{1}\right)$$

For any vector $v \in \mathbb{R}^{\mathcal{S}}$ and any scalar constant $c \in \mathbb{R}$, adding a constant vector $c\mathbf{1}$ does not change its span semi-norm:

$$\text{sp}(v + c\mathbf{1}) = \text{sp}(v)$$

This is because $\max_s(v(s)+c) - \min_s(v(s)+c) = \max_s v(s) + c - \min_s v(s) - c = \max_s v(s) - \min_s v(s) = \text{sp}(v)$.

Since $g^*\mathbf{1}$ is a constant vector across all states, we apply the above property of the span semi-norm ($\text{sp}(u + c\mathbf{1}) = \text{sp}(u)$ with $u = P(h^k - h^*)$ and $c = g^*$):

$$\text{sp}\left(P(h^k - h^*) + g^*\mathbf{1}\right) = \text{sp}\left(P(h^k - h^*)\right)$$

Thus, we get:

$$\mathrm{sp}(h^{k+1} - h^*) = \mathrm{sp}\left(P(h^k - h^*)\right)$$

By the problem statement, the transition operator P satisfies $\mathrm{sp}(Pv) \leq \gamma\,\mathrm{sp}(v)$ for all vectors v, where $\gamma \in [0, 1)$.

Setting $v = h^k - h^*$, we get:

$$\mathrm{sp}\left(P(h^k - h^*)\right) \leq \gamma\,\mathrm{sp}(h^k - h^*)$$

Through this, we get the contraction inequality:

$$\mathrm{sp}(h^{k+1} - h^*) \leq \gamma\,\mathrm{sp}(h^k - h^*)$$

For the bound for any arbitrary iteration k, we apply the inequality inductively.

For $k = 1$: $\mathrm{sp}(h^1 - h^*) \leq \gamma\,\mathrm{sp}(h^0 - h^*)$

For $k = 2$: $\mathrm{sp}(h^2 - h^*) \leq \gamma\,\mathrm{sp}(h^1 - h^*) \leq \gamma^2\,\mathrm{sp}(h^0 - h^*)$

Recursively for k steps, we get:

$$\mathrm{sp}(h^k - h^*) \leq \gamma^k\,\mathrm{sp}(h^0 - h^*)$$

8.2 (b)

Let us define the difference vector $v = h - h^*$.

By the premise of the problem, we are given that:

$$\mathrm{sp}(h - h^*) = \mathrm{sp}(v) \leq \epsilon$$

Let M and m denote the maximum and minimum values of the vector v over all states in $\mathcal{S}$, respectively:

$$M = \max_{s \in \mathcal{S}} v(s), \quad m = \min_{s \in \mathcal{S}} v(s)$$

By definition of the span semi-norm above, we have:

$$M - m \leq \epsilon$$

For every state $s \in \mathcal{S}$, the components of v naturally satisfy the inequality:

$$m \leq v(s) \leq M$$

To decompose v, we choose the scalar c to be the midpoint between the maximum and minimum values of v, so that it is equidistant from both the extremes:

$$c = \frac{M + m}{2}$$

Next, we define the vector $e \in \mathbb{R}^{\mathcal{S}}$ component-wise as the deviation of each element of v from this midpoint c:

$$e(s) = v(s) - c \quad \text{for all } s \in \mathcal{S}$$

Rewriting this relation gives us $v(s) = c + e(s)$, which in vector form translates to:

$$h - h^* = c1 + e \implies h = h^* + c1 + e$$

where 1 is the vector of all ones.

We will now bound the infinity norm of e, defined as $\|e\|_\infty = \max_{s\in\mathcal{S}} |e(s)|$.

For any state $s \in \mathcal{S}$, substituting $c = \frac{M+m}{2}$ into the expression for $e(s)$, we get:

$$e(s) = v(s) - \frac{M+m}{2}$$

Using the bounds $m \leq v(s) \leq M$, we can bound $e(s)$ from both max and min:

$$m - \frac{M+m}{2} \leq e(s) \leq M - \frac{M+m}{2}$$

$$\frac{2m - M - m}{2} \leq e(s) \leq \frac{2M - M - m}{2}$$

$$\frac{m - M}{2} \leq e(s) \leq \frac{M - m}{2}$$

From definition, since $M - m = \text{sp}(v)$, we can rewrite this interval as:

$$-\frac{\text{sp}(v)}{2} \leq e(s) \leq \frac{\text{sp}(v)}{2}$$

Taking the absolute value gives:

$$|e(s)| \leq \frac{\text{sp}(v)}{2}$$

Since this holds for all $s \in \mathcal{S}$, taking the maximum over all states gives:

$$\|e\|_\infty = \max_{s\in\mathcal{S}} |e(s)| \leq \frac{\text{sp}(v)}{2}$$

Given that $\text{sp}(v) = \text{sp}(h - h^*) \leq \epsilon$, we get at the desired bound as:

$$\|e\|_\infty \leq \frac{\epsilon}{2}$$

8.3 (c)

As defined in the question, the sequence of estimated gains at iteration k is given by:

$$g^k = L(h^k) - h^k = r + Ph^k - h^k$$

At iteration k, let $\epsilon_k = \text{sp}(h^k - h^*)$.

From previous question, we know that:

$$\epsilon_k \leq \gamma^k \,\text{sp}(h^0 - h^*)$$

Applying the result of previous part to h^k, there exists a scalar c_k and a residual vector e^k such that:

$$h^k = h^* + c_k 1 + e^k$$

where the infinity norm of e^k is bounded by:

$$\|e^k\|_\infty \leq \frac{\epsilon_k}{2} \leq \frac{\gamma^k \operatorname{sp}(h^0 - h^*)}{2}$$

Substituting $h^k = h^* + c_k 1 + e^k$ into the expression for g^k, we get:

$$g^k = r + P(h^* + c_k 1 + e^k) - (h^* + c_k 1 + e^k)$$

Using the linearity of matrix-vector multiplication and the property that P is a stochastic transition matrix (meaning its row sums equal 1, so $P1 = 1$), along with the fact that c_k is a scalar ($P(c_k 1) = c_k P1 = c_k 1$):

$$P(h^* + c_k 1 + e^k) = Ph^* + c_k P1 + Pe^k = Ph^* + c_k 1 + Pe^k$$

Substituting this back into g^k:

$$g^k = r + Ph^* + c_k 1 + Pe^k - h^* - c_k 1 - e^k$$

Grouping terms and canceling the $c_k 1$ terms:

$$g^k = (r + Ph^* - h^*) + Pe^k - e^k$$

From the given condition equation in the problem ($g^* 1 + h^* = r + Ph^*$), we have:

$$r + Ph^* - h^* = g^* 1$$

Substituting this equivalence into our expression for g^k:

$$g^k = g^* 1 + Pe^k - e^k$$

Now re-arranging terms to isolate the error vector $g^k - g^* 1$:

$$g^k - g^* 1 = Pe^k - e^k$$

Taking the infinity norm on both sides and applying the triangle inequality:

$$\|g^k - g^* 1\|_\infty = \|Pe^k - e^k\|_\infty \leq \|Pe^k\|_\infty + \|e^k\|_\infty$$

Since P is a transition probability matrix (non-negative entries with rows summing to 1), its induced infinity norm is $\|P\|_\infty = 1$. Therefore,

$$\|Pe^k\|_\infty \leq \|P\|_\infty \|e^k\|_\infty = 1 \cdot \|e^k\|_\infty = \|e^k\|_\infty$$

Combining these gives:

$$\|g^k - g^* 1\|_\infty \leq \|e^k\|_\infty + \|e^k\|_\infty = 2\|e^k\|_\infty$$

Substituting the bound on $\|e^k\|_\infty$ from previous step ($\|e^k\|_\infty \leq \frac{1}{2}\gamma^k \operatorname{sp}(h^0 - h^*)$):

$$\|g^k - g^* 1\|_\infty \leq 2\left(\frac{1}{2}\gamma^k \operatorname{sp}(h^0 - h^*)\right) = \gamma^k \operatorname{sp}(h^0 - h^*)$$

Since $\gamma^k \leq 2\gamma^k$ (for $\gamma \in [0, 1)$ and $k \geq 0$), this also satisfies:

$$\|g^k - g^* 1\|_\infty \leq 2\operatorname{sp}(h^0 - h^*)\gamma^k$$

Reference to different lecture notes used for this homework:

- Notes 1
- Notes 2
- Notes 3
- Notes 4
- Notes 5
- Notes 6
- Notes 7
- Notes 8
- Notes 9
- Notes 10
- Notes 11
- Notes 12
- Notes 13
- Notes 14
- Notes 15
- Notes 16
- Notes 17
- Notes 18
- Notes 19

- Notes 20
- Notes 21
- Notes 22